

Numerical Verification and Proof-Guided Correction of a False Conjecture

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Abstract

1 Numerical search has become a routine instrument of mathematical research: a conjecture
2 that survives thousands of candidate counterexamples is often written down as a
3 result. We report a case in which this practice failed, and in which the failure was found
4 only because a human proof attempt and a machine search corrected each other. In a
5 theory of what it costs a decision-maker to learn how its own actions change the data it
6 later learns from, a lower bound says this cost cannot fall below a fixed floor, and we
7 asked whether an adversary who knows the functional form of the response could beat it.
8 An adversarial search over thousands of experimental designs said no, with a best ratio of
9 1.005, and we recorded the floor as structure-proof. The proof we then wrote established
10 the bound only within two curvature lengths of the operating point. Searching exactly
11 where the proof stopped produced a counterexample: a four-point design with one far
12 probe of weight 3×10^{-4} , attaining ratio 0.8517. The original verdict was a search
13 artifact; the corrected result is two-sided and exact. The episode sits among eleven such
14 corrections recorded in the same project.

15 1 Introduction

16 Machines now participate across the mathematical research loop: they suggest conjectures [Davies
17 et al., 2021], search for counterexamples [Wagner, 2021, Romera-Paredes et al., 2024], and prove
18 theorems [Trinh et al., 2024, Hubert et al., 2025]. Most reported successes run in one direction: the
19 machine proposes or refutes, and a person proves. The failure direction, in which a numerical verdict
20 is wrong and the record of how it went wrong is preserved, is rarely documented.

21 This paper documents one such failure completely, inside a body of theory about the cost of self-
22 knowledge in performative prediction (Section 2). A machine search supported a claim. A human
23 proof attempt could establish the claim only on a region of the design space, and it named the
24 boundary of that region. A second machine search, restricted to the outside of that region, refuted
25 the claim. Neither half of the loop found the result alone. We then place it among ten more **pivots**,
26 measurement-forced changes to claims already written down, from the same project. This is not an
27 autonomous agent: people wrote the proofs; the machine checked, searched, and verified (Section 3).

28 **Contributions.** (1) A reproducible account of one proof-and-search correction cycle, ending in a
29 frozen counterexample and an exact theorem (Section 4, Theorem 2). (2) Ten further pivots from the
30 same project and an assessment of what the sample supports (Sections 4 and 5). (3) The infrastructure
31 that made the correction detectable, shipped as an anonymized repository. Related work, formal
32 statements, proofs, further verification, and the reference list are in the supplementary material.

33 2 Setting: the cost of self-knowledge

34 The object at stake in Section 4 is a *lower bound*, or floor: a statement that no strategy, however
35 clever, can do better than a stated amount, so a single design that beats it falsifies it.

36 Consider a decision-maker whose deployed action changes the data it will later retrain on. A dealer
 37 who quotes a wide spread sees different order flow than one who quotes a tight spread, so the flow
 38 it must learn to price is partly its own doing [Nagarajan and Ashok, 2026]. This is **performative**
 39 **prediction** [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023]: retraining becomes a fixed-
 40 point iteration, and it converges to a point that is not the value-maximizing one. Algorithms that
 41 close this gap [Izzo et al., 2021, Miller et al., 2021, Drusvyatskiy and Xiao, 2023] need an estimate
 42 of how the world responds to the action, and they obtain it by *exploring*, that is, by deploying actions
 43 away from the current best guess and observing what changes. Those exploratory actions are real
 44 deployments, so they forgo value; the theory prices that loss.

45 **Notation.** Let h denote the scalar action, h^* the operating point the decision-maker would choose
 46 with full knowledge, and $d_t = h_t - h^*$ the deviation deployed at round t . Let ε be the *response slope*,
 47 the quantity to be estimated; σ^2 the observation noise variance; Φ the value function; and γ_{PO} its
 48 curvature at h^* , the rate at which value is lost away from h^* . The cost of exploration is the value
 49 forgone, $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$. Labels such as T2 are theorem-register identifiers (formal
 50 statements: supplement, Section A).

51 **Theorem 1** (Exchange rate; T2). *Along any stationary exploration policy, for the least-squares slope*
 52 *estimator, $\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + o(1))$, independent of the exploration amplitude, shape,*
 53 *schedule, and of the contraction strength of the loop.*

54 Precision and cost trade off at a fixed exchange rate: exploring more reduces the variance of the
 55 estimate and raises the cost by exactly offsetting amounts, so their product is a constant. The reason
 56 is a cancellation: the information a design yields and the value it forgoes both grow with the same
 57 quantity, the spread of the deployed actions around h^* , so their ratio is fixed. Figure 1 (left) shows
 58 the measured product lying on the predicted constant. Further results extend the floor to every policy
 59 and estimator (T3, T4), and the theory was instantiated on the corporate-bond market of Nagarajan
 60 and Ashok [2026] and a second economy (Table 1; supplement, Section A).

61 3 Methodology: the verification loop

62 Every claim in the project passes through the following procedure, listed in the order it occurred for
 63 the pivot of Section 4. (1) **Register the claim** with an identifier and a proof status (proved, proved for
 64 a stated class, conditional on a hypothesis, or numerically supported). (2) **Preregister a numerical**
 65 **check:** a closed form is compared to a simulation at a tolerance fixed before the run. (3) **Search**
 66 **adversarially:** for a claim of the form “every design satisfies an inequality”, a random and locally
 67 refined search over designs looks for a violation. (4) **Attempt a proof:** if the proof succeeds only on
 68 a sub-region, its boundary is recorded and the search of step 3 is re-run restricted to the unreached
 69 region. (5) **Verify and record:** a candidate counterexample is frozen, recomputed with `mpmath` at
 70 50-digit precision, and added to the suite that runs on every push.

71 Steps 2, 3, the re-run in 4, and the verification in 5 are automated; every proof, every choice of what
 72 to conjecture, and the decision to search outside the proof’s reach were made by people.

73 4 Results: the structure-proofness pivot

74 **The question.** Theorem 1 concerns one estimator on one policy class, and the minimax floors T3
 75 and T4 extend it to all policies and estimators. Both leave open the following. The adversary is a
 76 hypothetical experimenter with the best possible information. Suppose this adversary knows the
 77 functional form of the response, not only its slope: concretely, that it is $\tau_\theta(h) = C_0 + C_1 e^{-ch}$, an
 78 exponential with three unknown parameters, where c sets the curvature and $1/c$ is the length over
 79 which the response bends. Knowing the form is real information, so it might permit an experimental
 80 design that beats the floor.

81 **The initial search verdict.** A *design* μ is a set of probe locations with weights, the fraction of
 82 the exploration budget spent at each. Its Fisher information $M(\mu)$ measures what the probes reveal
 83 about the three parameters, and the Cramér–Rao bound $g^\top M(\mu)^{-1} g$ is the smallest variance any
 84 unbiased estimator of the slope can achieve under μ , where g is the gradient of the slope with respect

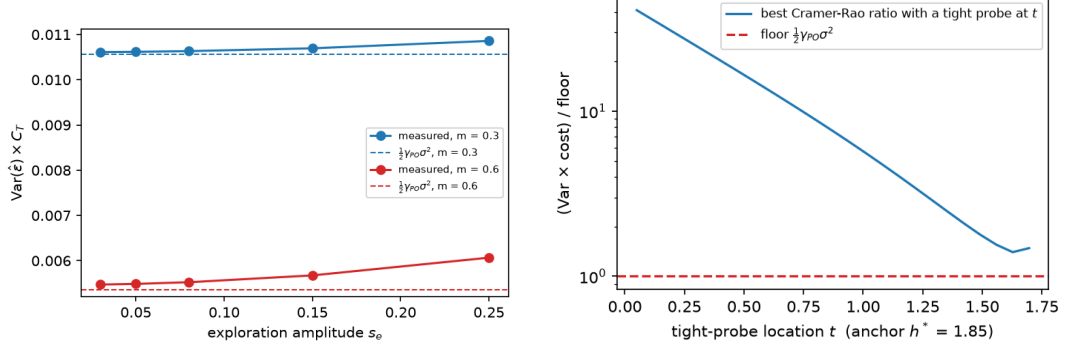


Figure 1: **Left:** the exchange rate of Theorem 1. Measured $\text{Var}(\hat{\varepsilon}) \times C_T$ against exploration amplitude at two contraction moduli m ; dashed lines are the predicted constant $\frac{1}{2}\gamma_{PO}\sigma^2$. **Right:** the search that produced the false verdict. Best Cramér–Rao ratio with one tight probe at location t , over the coarse weight grid with $w \geq 0.05$ (floor dashed). On this family, farther probes are monotonically worse, which is what we believed in general. The counterexample of Section 4 lies outside this family: probe weight 3×10^{-4} , inside an asymmetric cluster the search never generated.

to the parameters. The relevant quantity is this bound times the design’s cost, normalized by the floor, $R(\mu) = (g^\top M(\mu)^{-1}g) \int (h - h^*)^2 d\mu$, so that the floor survives structural knowledge exactly when $R(\mu) \geq 1$ for every μ . Four thousand random designs plus 2,500 local refinements returned a minimum of $R = 1.0050$, and a scan over designs pairing two near-anchor points with one distant probe found distant probes monotonically worse, reaching $R = 40.9$ at the grid’s edge (Figure 1, right). Two independent strategies agreed, and we recorded the claim: the floor is structure-proof, with best attained ratio 1.005. This claim was later shown to be false.

The proof attempt and its reach. The proof attempt reduces $R(\mu)$ to a one-dimensional problem. It rewrites $R(\mu)$ as a ratio: the cost of the design, $\int d^2 d\mu$ with $d(h) = h - h^*$, divided by the smallest weighted size $\int \phi^2 d\mu$ of any function ϕ built from the model’s sensitivities and normalized to unit slope at the anchor (supplement, Section B). It therefore suffices to exhibit one such ϕ that is everywhere no larger than $|d|$; then the ratio is at least one. There is a canonical candidate, the unique such function whose value, slope, and curvature at the anchor match those of d :

$$\phi_0(h) = \frac{2}{c} - e^{-c(h-h^*)} \left(\frac{2}{c} + h - h^* \right).$$

Writing $x = h - h^*$, the required domination $|\phi_0| \leq |x|$ holds for every $x \geq -2/c$, with equality only at $x = 0$ and $x = -2/c$; this is an elementary argument (supplement, Section B). It is also the whole story: for $x < -2/c$ the domination is not merely unproved but false, and it fails exponentially. The proof therefore established the floor for every design supported on the reach $[h^* - 2/c, \infty)$, from two curvature lengths below the anchor upward, and it identified the boundary exactly.

Targeted search and the witness. The failed step is not a gap in the argument but a location. We re-ran the adversarial search twice. *Inside the reach*, a hardened search (three runs of 30,000 random designs plus 10,000 refinements) pushed the minimum from 1.0050 down to $R = 1.00054$, consistent with the proof and with an infimum of exactly 1 that is approached but never attained. *Below the reach*, the same search found $R = 0.85164$. The witness is a four-point design with support $\{1.8726, 1.8665, 1.3073, 0.0500\}$ and weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$, in the market of Nagarajan and Ashok [2026] with $c = 1.5$ and anchor $h^* = 1.8461$. Every support point except the one at 0.05 lies inside the reach; the single below-reach probe of weight 3×10^{-4} is what breaks the floor. We froze the design and recomputed its ratio at 50-digit precision with `mpmath`, obtaining $0.851650\dots$; the check now runs in continuous integration on every push.

Why the first search missed it. First, the scan’s weight grid never went below 0.05, and the violating probe carries weight 3×10^{-4} . Second, the random search did generate distant probes, but always alongside a symmetric near-anchor pair, and such symmetric families never break the floor (their ratio approaches 1 from above as the probe weight vanishes); the violation requires

Table 1: Rows of the preregistered check harness that the text relies on (all 36 rows in the supplement; the drift row is a deliberately out-of-scope cell, reported rather than excused).

Claim (register)	Measured	Predicted	Verdict
Exchange rate, $m = 0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m = 0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	+16% over rate	out of scope	drift
Feedback floor, $m = 0.3/0.6/0.85$ (E1)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1 - m^2)$	pass
Nonparametric MSE at (B, T) optimum (T7, E5)	0.0132	0.0123	pass
Certified agent reaches h_{PO} ; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Real-data calibration port, 10 cells	10/10 reproduce	within 0.8%	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	$\geq 1; < 1$	pass

the asymmetric cluster. The earlier verdict was therefore not noise or a coding error but a correct summary of a search space that excluded the answer.

Theorem 2 (Structure-proofness, exact reach; T9). *(i) Every estimable design supported in $[h^* - 2/c, \infty)$ satisfies $R(\mu) > 1$ strictly, and $\inf_{\mu} R(\mu) = 1$, approached but not attained. Within reach, knowing the family cannot beat the floor. (ii) The reach is exact: the four-point witness above attains $R = 0.8517$.*

Interpretation. Structure-proofness is an ignorance-of-curvature phenomenon with an exactly known extent: a design that reaches far enough below the anchor learns the curvature parameter c cheaply. The floor holds because the experimenter does not know how sharply the response bends and must pay to find out; a probe placed far below the anchor, where the exponential is large, answers that question almost for free. With c known a priori, the floor breaks outright and keeps falling ($R = 0.3002$ at probe $t = 1.0$ and 0.0348 , still decreasing, at $t = 0.001$). And the reach contains the trust region of the certified agent of Table 1 ($|d_t| \leq 0.8 < 2/c = 1.333$), so the constraint that keeps the agent safe is the one that keeps its lower bound valid. Scope: $R(\mu)$ and the floor are defined under the local-quadratic cost model (assumption A1), and under the true market cost the same witness has ratio 1.0112; the counterexample fixes the bound’s scope, not a way to beat the market.

Further pivots. Three of the eleven pivots repeat the pattern above at smaller scale. **E1:** a saturation cap derived without noise failed by a factor of four in the noisy pipeline; the missing term was that retraining on noisy observations adds a stationary excitation floor of $(\sigma/\gamma)^2/(1 - m^2)$ per step, itself now verified to within 0.2 to 6.4 percent (Table 1). **E5:** an injected misspecification ae^{-2ch} had decayed to about 10^{-3} by the operating point and so was correctly invisible to the fit; a linear leak the family cannot represent produced the predicted crossover (Table 1). **E7:** a claimed threefold advantage for shaped exploration collapsed to parity once the comparison arm was funded with the same energy; the corrected claim is sharper: the identification cliff is a support-and-amplitude phenomenon, not a shape one. Each replacement was stronger than the claim it displaced.

5 Discussion, limitations, and conclusion

What the sample supports. On this project, numerical falsification was cheap and frequent, and it improved claims rather than only deleting them. It supports no claim about proof automation or other areas of mathematics. Unlike work in which a search refutes a conjecture directly, here the search had returned the wrong verdict, and a proof’s failure rather than a larger budget redirected it.

Infrastructure. A register lists all 26 results with proof status. A preregistered harness compares each measurement to a closed form at a preset tolerance; 36 rows pass and one out-of-scope cell is reported as drift (Table 1). We label the sharp constant $\frac{1}{2}$ in T4 as open and our own structure-proofness verdict as a search artifact: a record containing only successes cannot calibrate trust in the next verdict. It is independent of the mathematics and is what we recommend adopting.

Limitations. Theorem 1 is local: at finite exploration amplitude the identity is only approximate, and the deliberately out-of-scope saturating cell drifts by +16 percent. Which below-reach designs violate Theorem 2’s floor, and by how much, is open; and eleven pivots from one group are a sample, not a study.